NATIONAL UNIVERSITY OF SINGAPORE

ST5202 Applied Regression Analysis

Semester II, 2020-2021

Time allowed: 2 hours

INSTRUCTIONS TO CANDIDATES

1. This exam paper contains FOUR questions and NINE pages.
2. The exam is open book.
3. You are not allowed to use R for the computation. But any type of calculator can be used.
4. Your answers will be checked for similarities to the others. If high similarity is found, a disciplinary investigation will be launched. It might result in disciplinary actions against you.
5. Name your answer file in the following format:

   `last name_first name_matriculation number.pdf`.

   Example: `Chen_zehua_08235J.pdf`.
6. Your answer files are to be submitted to the module's LumiNUS folder: Files/Final Exam Submission. The folder opens at 6:45 pm and closes at 7:30 pm, May 3, 2021.
7. To avoid possible site crash, submit your answer files as soon as possible after you finish your writing, do not wait until the last moment. Your submission will not be accepted after the folder close time.

1. Three brands of batteries are under study. It is expected that the average life (in weeks) of Brand 2 is shorter than the average life of Brand 1 and that the average life of brand 1 is shorter than the average life of Brand 3. To verify these expectations, five batteries of each brand are tested for their lives. The resultant data is fitted by the model:

$$y = \beta_0 + \beta_2 * \texttt{brand2} + \beta_3 * \texttt{brand3} + \epsilon,$$

where y is the life of batteries, `brand2` and `brand3` are dummy variables for factor brand defined in the conventional way. The fitted results are as follows.

```
lm(formula = y ~ brand)
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   95.200      1.766  53.896  1.1e-15 ***
brand2       -15.800      2.498  -6.325  3.8e-05 ***
brand3         5.200      2.498   2.082   0.0594 .
```

(i) Test the expectations mentioned above by controlling the overall type I error rate at level $\alpha = 0.05$ with an appropriate crierion. State the hypotheses in terms of the parameters in the model. Compute the test statistics. State the decision rule and draw your conclusion.

(ii) Suppose we just want to use a criterion to control the overall type I error rate for the tests in (i).

(iia) Among Sheffe, Tukey and Bonferroni's criteria, which criterion can be used? Give your reasons.

(iib) For the criteria which can be used to control the overall type I error rate, obtain the critical value of each of those criteria, making adjustment for one-sided tests. Judging by the critical values, which criterion is the best? Why?

The following are some quantiles for your reference. t-distribution: $t_{12}(0.05) = 1.7823, t_{12}(0.025) = 2.1788, t_{12}(0.0125) = 2.560$. F-distribution: $F_{2,12}(0.025) = 5.0959, F_{2,12}(0.05) = 3.8853, F_{2,12}(0.1) = 2.8068$. Studentized range distribution: $q_{3,12}(0.025) = 4.325, q_{3,12}(0.05) = 3.773$, $q_{3,12}(0.1) = 3.204$.

2. In a clinical trial, three medical treatments are randomly assigned to 15 patients such that each treatment is applied to 5 patients. In addition to the end-point measurement y, there is available a measurement of pre-trial health status x which is thought to have potential effect on the end-point measurement. To allow the investigation of possible interaction of x and the treatments, the following interaction model is considered:

$$y = \beta_0 + \beta_2 T_2 + \beta_3 T_3 + \alpha z + \gamma_1 T_2 z + \gamma_2 T_3 z + \epsilon,$$

where $z = x - \bar{x}$, T_2 and T_3 are the dummy variables for the treatments defined in the conventional way. The fitted results are given below:

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 14.71190    0.99305  14.815 1.26e-07 ***
T2           4.28521    1.39781   3.066   0.0134 *
T3          -3.58306    1.41507  -2.532   0.0321 *
z            1.52929    0.74997   2.039   0.0719 .
T2:z         0.34951    0.99102   0.353   0.7325
T3:z         0.08494    1.20679   0.070   0.9454

Variance-covariance matrix:
          T2      T3       z    T2:z    T3:z
T2    1.9539  0.9861 -0.1046  0.1254  0.1046
T3    0.9861  2.0024 -0.1046  0.1046 -0.1060
z    -0.1046 -0.1046  0.5625 -0.5625 -0.5625
T2:z  0.1254  0.1046 -0.5625  0.9821  0.5625
T3:z  0.1046 -0.1060 -0.5625  0.5625  1.4563
```

(i) Test whether or not the interaction between x and the treatments is significant at level 0.05. State the hypotheses in terms of the parameters in the model. Compute the test statistic. State the decision rule and draw your conclusion. [You might need to use the following formula: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -c \\ -b & a \end{pmatrix}$.]

(ii) As a general exploring, the clinician wants to compare the three treatments when the pre-trial health status x is held at its average value. Let μ_1, μ_2 and μ_3 denote the mean response to the three treatments while x is held at the average value. A few contrasts are identified from the fitted results which includes: $\mu_2 - \mu_1$, $\mu_2 - (\mu_1 + \mu_3)/2$. Construct 95% simultaneous confidence intervals for these two contrasts using an appropriate multiple comparison criterion.

(iii) If the clinician is only interested in the pairwise comparison while x is held at the average value, conduct the tests for the three pairs at level 0.05. State the hypotheses in terms of the parameters of the interaction model. Give the values of the test statistics. Give the critical values of the tests. State the decision rule and draw your conclusion.

The following are some quantiles for your reference. F-distribution: $F_{2,9}(0.05) = 4.2565, F_{3,9}(0.05) = 3.8625, F_{2,10}(0.05) = 4.1028$, $F_{3,10}(0.05) = 3.7083$. Studentized range distribution: $q_{3,9}(0.05) = 3.948, q_{2,9}(0.05) = 3.119$.

3. In a market study, data are collected for sales of roof shingles from 26 districts. The data consists of the following variables: y – sales in thousand dollars, x_1 – proportional expenditure in thousand dollars, x_2 – number of active accounts, x_3 – number of competing brands and x_4 – district potential.

   (i) To build up a model for the data, a forward selection procedure is carried out. The following shows the result of one step of the forward procedure with some information missing:

```
Step:  AIC=276.73
y ~ x3
       Df       SSR   SSE
+ x2    1     52502  2210
+ x4    1     10639 44073
<none>              54712
+ x1    1      2887 51825
```

   Based on the above result, answer the following questions.

   (ia) Which variable is the candidate variable to be selected at this step? Give your reason.

   (ib) Can the candidate variable be selected by the criterion AIC?

   (ii) The final model determined for the data is $y = \beta_0 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$. The fitted results are given below.

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 186.6940    12.2587   15.23 1.66e-13
x2            3.4081     0.1458   23.37  < 2e-16
x3          -21.1930     0.8028  -26.40  < 2e-16
Residual standard error: 9.803 on 23 degrees of freedom
F-statistic: 918.3 on 2 and 23 DF,  p-value: < 2.2e-16
```

   In addition, $\sum_{i=1}^{n}(x_{2i}-\bar{x}_2)^2 = 5051.3846$, $\sum_{i=1}^{n}(x_{3i}-\bar{x}_3)^2 = 166.6538$.

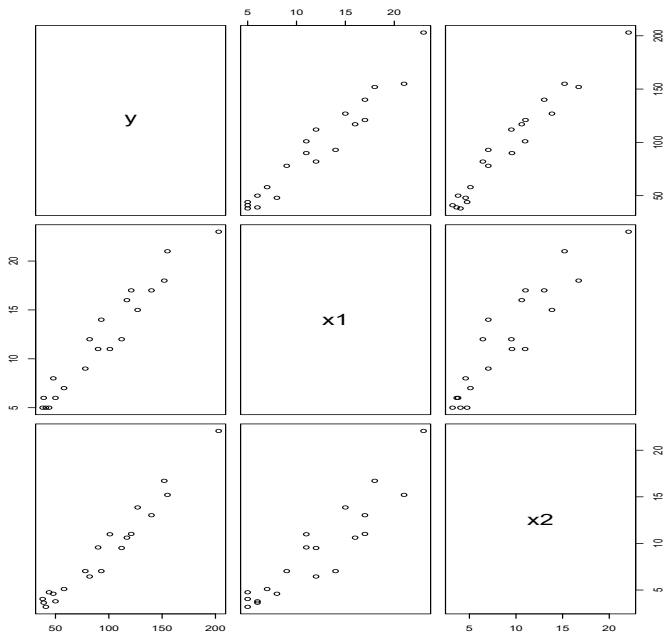
   (iia) Compute the variance inflation factor (VIF) for x_2 and x_3. Verify that the two VIF's are the same (they might differ a little due to rounding errors).

(iib) In general, if there are only two predictor variables in a regression model, the VIF of the two predictor variables are exactly the same. Explain why.

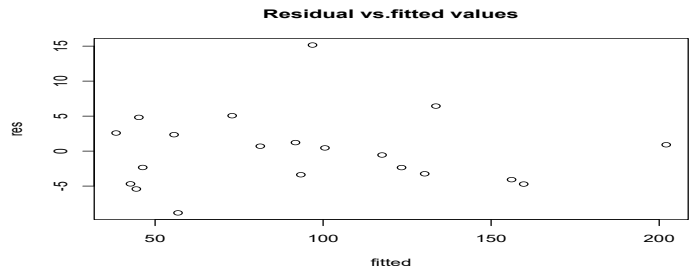
(iii) Test the significance of the model given in (ii) at level 0.05. State the hypotheses. Give the test statistic and its value. State the decision rule and draw your conclusion.

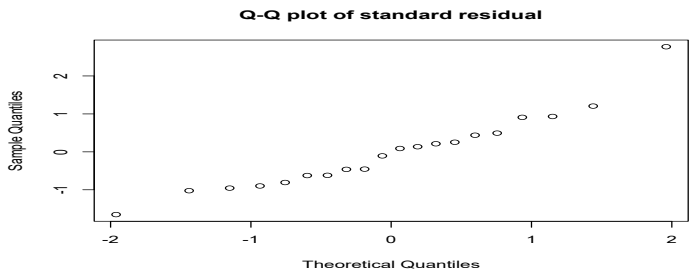
4. A sample of 20 chemical shipments were taken and the following data were collected: the time in minutes to handle shipments (y), the number of drums in shipment (x_1), the total weight of shipment in pounds (x_2). The following study is to establish a relationship between y and x_1, x_2.

   (i) The scatter plot matrix is given below. What do the scatter plots suggest about the nature of the relationship between y and each of x_1 and x_2? Is there any evidence of multicollinearity problem? Give your reasons. Suggest a proper regression model to entertain.

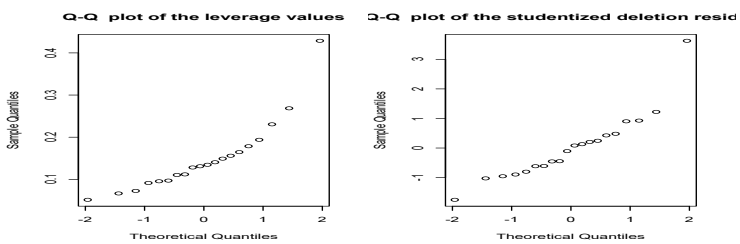


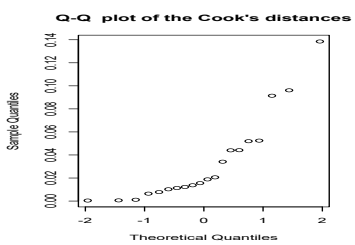
   (ii) The regression model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$ is fitted to the data. The plot of residuals versus fitted values and the Q-Q plot of the standardized residuals are given below. Do the plots exhibit any evidence of violation of homogeneity and normality assumptions? Are there any evidence of outliers? Provide your reasons.





(iii) The Q-Q plots of the hat values, standardized deletion residuals and the values of Cook's distance are given below.





Identify roughly the numbers of possible outliers with high leverage, high in-consistency and high influence respectively. What is your next step for the diagnostics? What should you do with these outliers?

(iv) The model with three additional variables: `u`, `v` and `w`, is fitted, where `u` is the dummy variable for observation 12, i.e., it equals 1 for observation 12 and 0 otherwise, similarly, `v` and `w` are dummy variables for observation 15 and 18 respectively. The fitted results are as follows:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.6678     2.2429   1.189  0.25406
x1            4.0858     0.4404   9.278 2.34e-07 ***
x2            4.7329     0.4624  10.236 6.99e-08 ***
u            15.2933     3.8749   3.947  0.00146 **
v            -9.1251     3.9636  -2.302  0.03719 *
w            -5.4560     4.2984  -1.269  0.22503
```

If the family-wise error rate of declaring any of these three observations as significant outliers is controlled at level $\alpha = 0.05$, which of the observations are significant at this level? Give your argument.

(v) The value of the variance inflation factor of x_1 and x_2 is computed as 7.0275. (a) Do you think that the least square estimate of the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$ is reliable for the inference of the effect of x_1 on y? (b) If you want to build up a model for the prediction of y by using x_1 and x_2, what strategy do you adopt for fitting the model?

END OF PAPER